

Project Executable Files

Date	10 Jul 2026
Team ID	XXXXXX
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example similar) or	NA
6	Deployed application URL (if hosted)	NO

7	APK / Executable binary (if applicable for mobile/desktop apps)	
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

/Smart_Lender_AI

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|— app.py           - Main Flask application
|— train_model.py   - Machine Learning model training
|— utils.py         - Utility/helper functions
|— loan_data.csv     - Loan dataset
|— requirements.txt  - Python dependencies
|— README.md        - Project documentation
|— test_output.txt   - Test output results
|— test_err.txt      - Error log (testing)
  
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└— models/
|   └— model.pkl     - Trained ML model
|   └— encoder.pkl   - Label encoder
|   └— scaler.pkl    - Feature scaler
  
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└— templates/
|   └— index.html    - Home page
  
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| ├── predict.html    - Loan prediction page
| ├── history.html    - Prediction history
| ├── profile.html    - Applicant profile
| └── about.html      - About page

└── static/
    ├── css/          - Stylesheets
    ├── js/            - JavaScript files
    └── images/        - Images and icons

└── __pycache__/      - Python cache files
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Deployed (Runs on Local Host: http://127.0.0.1:5000)
Login Credentials (Demo)	Not Applicable (No Login Authentication Implemented)
Platform / Hosting Provider	Local Flask Development Server (Visual Studio Code)
Repository Link	https://github.com/anushaobulapuram/smart_lender_smart_bridge.git
Demo Video Link	https://drive.google.com/file/d/1SFsukLL6qTV93eniIDsgYuXsOK5jTuJl/view?usp=drivesdk

Step 4: Run Instructions

Pre-requisites:

- Python 3.10 or above
- Visual Studio Code
- Git (Optional)
- Required Python libraries (requirements.txt)

Step 1: Clone the repository git clone

https://github.com/anushaobulapuram/smart_lender_smart_bridge.git

Step 2: Navigate to the project folder

`cd smart_lender_smart_bridge`

Step 3: Install the required dependencies pip

`install -r requirements.txt`

Step 4: Train the Machine Learning model (if model files are not available)

`python train_model.py`

Step 5: Run the Flask application

`python app.py`

Step 6: Open the application in your web browser

<http://127.0.0.1:5000>

Step 7: Enter the applicant details and click "Predict Loan" to view the loan eligibility prediction, applicant profile, prediction history, and download the PDF report.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application runs only on a local system and is not deployed online.	Runs successfully on the local Flask server; cloud deployment can be added in the future.

2	Prediction accuracy depends on the quality and completeness of the dataset.	A cleaned and preprocessed dataset was used to improve prediction accuracy.
3	The system currently supports only a single loan prediction model and does not integrate with real-time banking databases.	Future enhancement includes multiple loan models and real-time database/API integration.